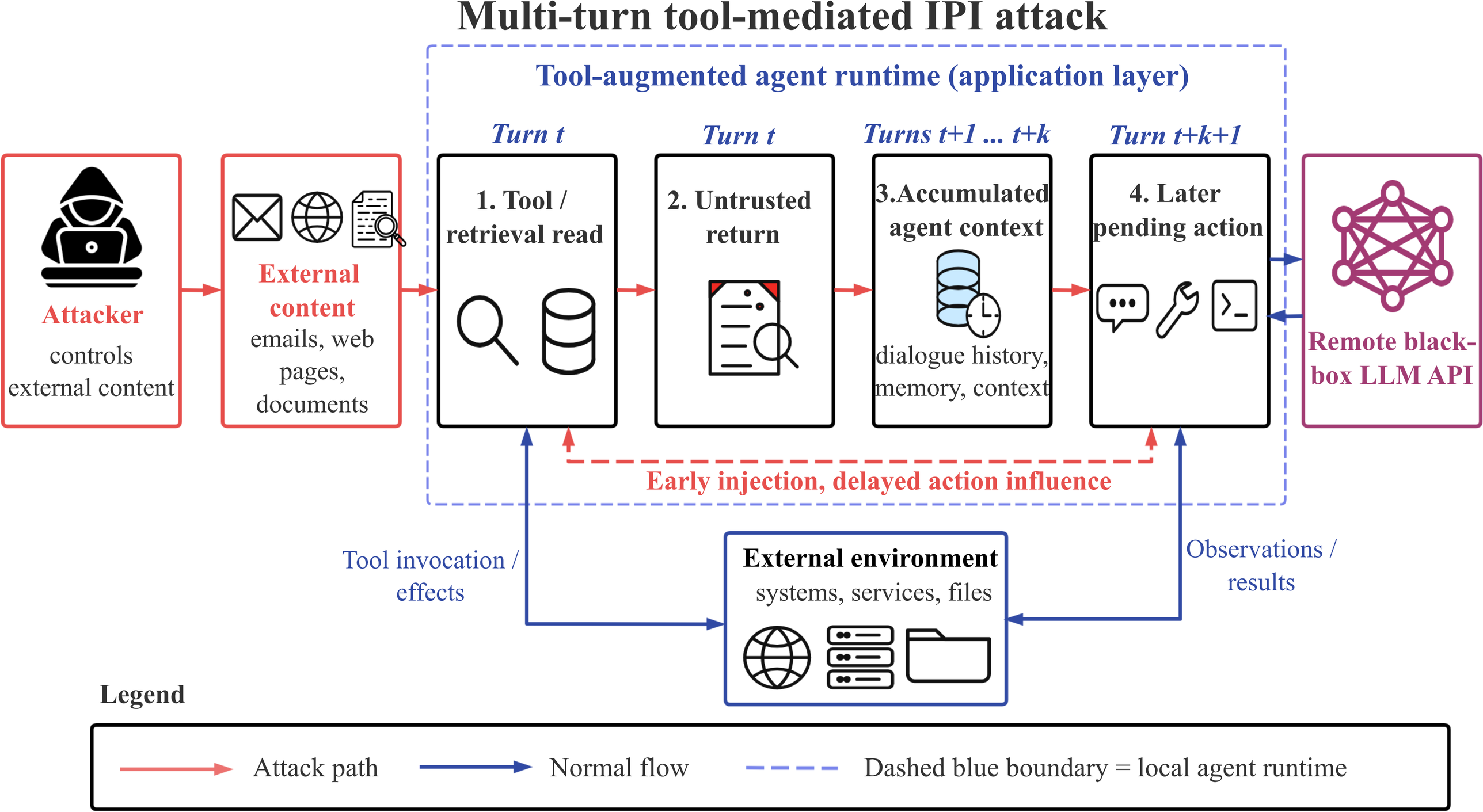

Multi-turn tool-mediated IPI attack
Tool-augmented agent runtime (application layer)
Turn t
Turn t
Turns t+1 ... t+k
Turn t+k+1
Attacker
controls external content
External content
emails, web pages, documents
1. Tool / retrieval read
2. Untrusted return
3.Accumulated agent context
dialogue history, memory, context
4. Later pending action
Remote black-box LLM API
Early injection, delayed action influence
Tool invocation / effects
External environment
systems, services, files
Observations / results
Legend
Attack path
Normal flow
Dashed blue boundary = local agent runtime